Analysis for the Beveridge Curve

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The Beveridge curve usually shows an inverse relationship, meaning that as job vacancies rise, unemployment typically falls. But this isn't always the case, especially when structural issues come into play. For example, companies might hire from the existing workforce instead of unemployed people. Labor market frictions, like skill mismatches, tech changes, or education gaps, can push the curve outward. A good example of this was the 2008 financial crisis, where long-term unemployment caused the curve to shift outward. If these structural problems are addressed, the shift could reverse. Economic growth usually brings more job openings, but if skilled workers are in short supply, the curve may still shift outward, causing higher unemployment even as vacancies rise. Inflation, meanwhile, tends to drive up interest rates, reducing labor demand and shifting the curve inward. However, inflation can also increase vacancies in sectors like construction, even if overall unemployment stays high. New businesses can also help by boosting job creation and competition, which may reduce unemployment as firms compete for workers.

In my analysis, I started by using an ARIMA(1, 1, 0) model in Stata to predict shifts along the Beveridge curve. The results showed a small initial decrease followed by an increase in unemployment, which contradicted my expectation that unemployment would decrease in the short run due to protectionist policies reducing the labor supply. Since firms would likely hire from the existing workforce, the failure of the model to capture this dynamic suggested that the ARIMA model, which relies on historical data, may not be appropriate for this context. I then checked the forecasts with a VAR model, but the results still indicated rising unemployment, which raised further concerns. Afterward, I used a linear multivariate regression model, based on forecasts from key macroeconomic indicators, which predicted a decline in unemployment. However, the extent of the decline seemed too steep, which pointed to potential misspecification of the model or oversensitivity to certain variables.

Sectoral analysis showed that unemployment should decrease, as growth trends in most sectors aligned with this expectation. Ideally, the results should have shown an initial decline in unemployment, followed by stagnation or modest increases due to labor market frictions, such as the service-manufacturing bias. Future research could refine this analysis by using models like those of Hsieh and Klenow (2010), which take protectionist policies into account, or by focusing on sector-specific labor market frictions. Unfortunately, due to time constraints and limited expertise in these more advanced models, further refinement wasn't possible.

Forcasting the Beveridge Curve

Zergham Hayder Khan

Introduction

First I investigate the pairwise correlations among GDP, employment, and profits across various economic sectors. The aim is to evaluate the extent to which these indicators are interrelated and the degree to which their co-movement aligns with theoretical expectations.

Regression Results(STATA)

1. Relationship Between Unemployment and Job Openings

I first estimated the relationship between Unemployment and Job Openings. The regression output is presented below:

Variable	**Coef.**	**Std. Err.**	**t**	**P¿—t—**	**95% Conf. Interval**
JobOpenings	-0.8898	0.0736	-12.10	0.000	[-1.0346, -0.7450]
_cons	9.0102	0.2836	31.77	0.000	[8.4519, 9.5686]

Interpretation:

- The coefficient for Job Openings is negative (-0.8898), indicating that as Job Openings increase, Unemployment tends to decrease.
- This relationship is statistically significant with a very low p-value ($p < 0.0001$).
- The constant term (_cons) is 9.0102, suggesting that when there are no job openings, Unemployment is predicted to be at approximately 9.01

Possible Explanations:

- A negative relationship between job openings and unemployment is in line with basic labor market theory: as more jobs become available, individuals are more likely to find employment, reducing the unemployment rate.
- The significant negative effect of job openings suggests that labor market tightness (i.e., more available jobs) is strongly associated with lower unemployment, consistent with expectations in a competitive job market.

2. Relationship Between Unemployment, Job Openings, and Quarterly GDP

Next, is estimated the relationship between Unemployment, Job Openings, and Quarterly GDP (in billions). The regression output is shown below:

Variable	Coef.	Std. Err.	t	P¿—t—	95% Conf. Interval
JobOpenings	-1.5548	0.2292	-6.78	0.000	[-2.0100, -1.0996]
QuarterlyGDPbillions	0.0002	0.00006	3.51	0.001	[0.000090, 0.000325]
_cons	7.8352	0.6032	12.99	0.000	[6.6373, 9.0331]

Interpretation:

- The coefficient for Job Openings is again negative (-1.5548), implying that more job openings are associated with a lower unemployment rate, as expected.
- The coefficient for Quarterly GDP (0.0002) is positive and statistically significant, meaning that as GDP increases, unemployment tends to decrease.
- The constant term (_cons) is 7.8352, suggesting that even when both Job Openings and GDP are zero, unemployment is predicted to be at 7.8352

Possible Explanations:

- The negative coefficient for Job Openings (-1.5548) reinforces the finding from the previous model that an increase in job openings leads to a reduction in unemployment.
- The positive relationship between GDP and unemployment may suggest that economic growth plays a crucial role in reducing unemployment, as higher GDP levels correspond to more economic activity and job creation.
- The statistically significant effect of Quarterly GDP supports the idea that a growing economy leads to lower unemployment, providing more opportunities for job creation.

Model Diagnostics

- **R-squared:** In both models, the R-squared values suggest that a portion of the variation in unemployment is explained by the predictors. In the first model, the R-squared value is 0.3400, and in the second model, it is 0.3924.
- **F-statistic:** The high F-statistics in both models (146.32 and 30.03) indicate that the models are statistically significant.
- **Root MSE:** The Root Mean Squared Error (RMSE) values of 1.6051 and 1.63 indicate that the models have reasonable predictive accuracy, although some error remains.

Sector-Wise Correlation Analysis

All Sectors (Overall Economy)

Indicator	GDP vs Employment	GDP vs Profits	Employment vs Profits
$All - GDP$	1.0000	0.9261	0.9432

Observation: These variables exhibit strong positive correlations. **Reasoning:** GDP reflects overall economic performance, which drives employment and profits. Their co-movement is expected as economic growth directly stimulates corporate activity and job creation.

Agriculture, Forestry, and Fishing

Indicator	GDP vs Employment	GDP vs Profits	Employment vs Prof
$Agriculture, Forestry, Fishing$	0.8280	0.7992	0.7761

Observation: Moderate to strong correlations are present. **Reasoning:** The sector's output is labor-intensive, so growth in GDP tends to increase employment and profitability, but natural and market factors introduce variability.

Mining

Indicator	GDP vs Employment	GDP vs Profits	Employment vs Profits
$Mining$	0.1944	0.0476	−0.0332

Observation: Correlations are weak or negligible. **Reasoning:** Mining profits depend more on global commodity prices than local GDP or employment, leading to a disconnect between these indicators.

Utilities

Indicator	GDP vs Employment	GDP vs Profits	Employment vs Profits
$Utilities$	0.1833	0.0429	0.2120

Observation: Correlations are weak. **Reasoning:** Utilities operate as essential services, with profits and employment less influenced by GDP fluctuations.

Construction

Indicator	GDP vs Employment	GDP vs Profits	Employment vs Profits
$Construction$	0.9311	0.6699	0.7135

Observation: Strong correlations, especially between GDP and employment. **Reasoning:** Construction activities expand with economic growth, creating jobs and boosting profits due to increased demand.

Manufacturing

Indicator	**GDP vs Employment**	**GDP vs Profits**	**Employment vs Profits**
$Manufacturing$	0.6472	0.1356	-0.0417

Observation: GDP and employment are moderately correlated, but profits show weak relationships. **Reasoning:** Manufacturing involves fixed costs, so profits can remain flat despite GDP growth and employment increases.

Retail Trade

Indicator	**GDP vs Employment**	**GDP vs Profits**	**Employment vs Profits**
$RetailTrade$	0.6176	0.8651	0.3110

Observation: GDP and profits are highly correlated, while employment has weaker connections. **Reasoning:** Retail profits respond quickly to economic growth, but employment may lag due to automation or seasonal patterns.

Education, Health, and Social Services

Indicator	**GDP vs Employment**	**GDP vs Profits**	**Employment vs Profits**
$EducationandHealth$	0.9765	0.4086	0.3308

Observation: GDP and employment are closely linked, while profits show moderate correlation. **Reasoning:** These sectors are employment-driven and often receive public funding, which stabilizes employment even when profits fluctuate.

Conclusion

The correlation patterns vary across sectors, reflecting the unique economic dynamics within each industry. While GDP and employment are often interrelated, the linkage to profits is more nuanced and depends on sector-specific factors.

ARIMA MODEL (STATA)

This study investigates the pairwise correlations among GDP, employment, and profits across various economic sectors. Additionally, it evaluates the stationarity of key economic variables using the Augmented Dickey-Fuller (ADF) test and fits an ARIMA model to model unemployment dynamics over time.

Unit Root Testing: Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test is applied to check for stationarity of the variables of interest: Unemployment, Job Openings, and Hires Rate. The null hypothesis of the ADF test is that the series has a unit root (i.e., it is non-stationary).

Results of Augmented Dickey-Fuller Test (ADF)

Variable	ADF Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Unemployment	-2.052	-3.458	-2.879	-2.570
Job Openings	-2.062	-3.458	-2.879	-2.570
Hires Rate	-1.549	-3.458	-2.879	-2.570

Interpretation: - The p-values for the ADF tests on the original variables (Unemployment: 0.2642, Job Openings: 0.2601, and Hires Rate: 0.5092) are all greater than 0.05, suggesting that none of these variables are stationary. Therefore, we reject the null hypothesis and conclude that they contain unit roots.

Stationarity After Differencing

To achieve stationarity, we difference the variables and perform the ADF test again on the differenced series.

Variable	ADF Statistic	1% Critical Value	5% Critical Value	10% Cri
Differenced Unemployment	-5.041	-3.458	-2.879	
Differenced Job Openings	-3.998	-3.458	-2.879	
Differenced Hires Rate	-5.222	-3.458	-2.879	

Interpretation: - After differencing, the p-values for the ADF tests are all less than 0.05 (p-values: 0.0000, 0.0014, and 0.0000), indicating that the differenced series for Unemployment, Job Openings, and Hires Rate are now stationary. This means that these variables no longer have a unit root and are suitable for time-series modeling.

Time-Series Modeling: ARIMA Model

Next, we fit a univariate ARIMA model to the differenced Unemployment series. The ARIMA model is chosen based on the autocorrelations in the series and is intended to model the dynamics of Unemployment over time.

ARIMA Model for Unemployment

```
arima Unemployment, ar(1) ma(1)
```

The ARIMA(1, 1, 1) model was fit to the differenced series of Unemployment, with the following results:

Variable	**Coefficient**	**Standard Error**	**z-value**	**P-value**
Constant	5.6115	1.6093	3.49	0.000
AR(1)	0.9335	0.0324	28.79	0.000
MA(1)	0.0676	0.0315	2.15	0.032

Interpretation: - The AR(1) coefficient is 0.9335, suggesting strong autocorrelation in the Unemployment series. A value close to 1 indicates that past values of Unemployment have a significant influence on future values. - The MA(1) coefficient is 0.0676, indicating a modest effect of past residuals (shocks) on future values. - The constant term (_cons) is 5.6115, which is statistically significant and indicates the long-run mean of the series after differencing.

Forecasts from the ARIMA Model

We now use the ARIMA model to predict future values of Unemployment. The predictions are based on the fitted model parameters.

```
predict Unemployment_hat, xb
```

The predicted values of Unemployment (denoted as `Unemployment_hat`) for the first 20 observations are as follows:

Observation	**Unemployment Prediction**
1	5.6115
2	4.0006
3	4.3073
4	4.2866
5	4.3881
6	4.4813
7	4.3749
8	4.5824
9	4.6684
10	4.9630
11	5.0432
12	5.3381
13	5.5184
14	5.7064
15	5.6937
16	5.6945
17	5.6945

18	5.8947
19	5.7811
20	5.7887

Interpretation: - The predicted Unemployment values show a steady decrease in the early predictions and stabilize around 5.7- The prediction results give an idea of the future path of Unemployment, assuming that the patterns captured by the ARIMA model hold in the future.

Conclusion

The results of the Augmented Dickey-Fuller tests indicate that the original time series for Unemployment, Job Openings, and Hires Rate are non-stationary. After differencing, all variables became stationary, allowing us to model Unemployment using an ARIMA model. The ARIMA model suggests a strong autocorrelation in Unemployment, with a significant contribution from both the autoregressive and moving average terms. The forecasted values indicate a stable downward trend in Unemployment.

Vector Autoregression (VAR) in Stata to check for lag effects are trends here can have persistence

Model Fit Statistics

Metric	Value
Log Likelihood	147.0393
AIC	-0.264
FPE	0.000155
HQIC	0.325
SBIC	1.204

Interpretation:

- The model has a reasonable fit as indicated by the log likelihood, AIC, and FPE.
- The low values of AIC and FPE suggest that the model is a good fit.

Equation Summary

Equation	Parms	RMSE	R-sq	Chi2	P ¿ Chi2
$d\_Unemployment$	37	0.658	0.2047	70.269	0.0005
$d\_JobOpenings$	37	0.151	0.4278	204.111	0.0000
$d\_HiresRate$	37	0.121	0.7630	878.995	0.0000

Interpretation:

- The $d\_HiresRate$ equation has the best fit with an R^2 of 0.763.

- $d\_JobOpenings$ has a moderate fit (R^2 of 0.4278).
- $d\_Unemployment$ has the lowest R^2 but still shows statistical significance.

Lag Effects for $d\_Unemployment$

Lag	Coef.	Std. Err.	P-value
L1	-0.185	0.066	0.005
L2	0.175	0.107	0.101
L6	-0.235	0.114	0.040

Interpretation:

- Lag 1 ($L1$) and Lag 6 ($L6$) show statistically significant effects on $d\_Unemployment$.
- Lag 2 ($L2$) is marginally insignificant ($p = 0.101$).

Lag Effects for $d\_JobOpenings$

Lag	Coef.	Std. Err.	P-value
L1	-0.849	0.264	0.001
L9	0.072	0.026	0.006
L12	-0.012	0.264	0.965

Interpretation:

- Lag 1 ($L1$) is highly significant with a strong negative coefficient.
- Lag 9 ($L9$) shows a positive and significant effect on $d\_JobOpenings$.
- Lag 12 ($L12$) is not significant.

Lag Effects for $d\_HiresRate$

Lag	Coef.	Std. Err.	P-value
L1	-1.415	0.363	0.000
L4	0.226	0.106	0.033
L12	0.106	0.051	0.036

Interpretation:

- Lag 1 ($L1$) has a significant negative impact on $d\_HiresRate$.
- Lag 4 ($L4$) and Lag 12 ($L12$) have smaller, but still statistically significant effects.

Model Selection Criteria

Model	AIC	BIC	df
Model	-72.079	328.573	111

Interpretation:

- AIC and BIC values suggest that the model is appropriately selected.

- The BIC is higher, suggesting possible overfitting; however, AIC indicates a good fit.

Results of VAR forecasts from R

Unemployment

Period	Forecast	Lower CI	Upper CI
1	4.27	1.42	7.12
2	4.60	1.12	8.07
3	5.14	1.45	8.83
4	4.54	0.66	8.41
5	4.89	0.81	8.96
6	5.05	0.85	9.25
7	4.92	0.58	9.26
8	5.06	0.61	9.51
9	5.02	0.50	9.54
10	5.05	0.45	9.64
11	5.12	0.47	9.78
12	5.15	0.43	9.86

Job Openings

Period	Forecast	Lower CI	Upper CI
1	4.74	4.21	5.27
2	4.62	3.81	5.43
3	4.52	3.49	5.55
4	4.61	3.33	5.89
5	4.55	3.02	6.08
6	4.55	2.81	6.28
7	4.57	2.66	6.48
8	4.53	2.46	6.60
9	4.51	2.30	6.72
10	4.51	2.19	6.83
11	4.49	2.07	6.91
12	4.47	1.97	6.97

GDP Growth Rate

Period	Forecast	Lower CI	Upper CI
1	0.30	−3.57	4.17
2	0.79	−3.87	5.44
3	0.04	−4.66	4.75
4	1.58	−3.15	6.30
5	0.08	−4.70	4.86
6	0.53	−4.30	5.36
7	0.85	−4.00	5.70
8	0.34	−4.52	5.20
9	0.81	−4.05	5.67
10	0.48	−4.39	5.34
11	0.53	−4.34	5.40
12	0.63	−4.24	5.50

Inflation Rate Difference

Period	Forecast	Lower CI	Upper CI
1	0.0033	−0.0118	0.0184
2	0.0017	−0.0165	0.0198
3	−0.0025	−0.0211	0.0161
4	0.0032	−0.0156	0.0221
5	−0.0036	−0.0226	0.0153
6	0.0001	−0.0193	0.0196
7	0.0014	−0.0182	0.0210
8	−0.0013	−0.0210	0.0184
9	0.0005	−0.0193	0.0202
10	−0.0008	−0.0205	0.0190
11	−0.0004	−0.0202	0.0193
12	0.0003	−0.0195	0.0201

1 Predicting using the Regression model

The following regression model is used to forecast unemployment:

$$\begin{aligned}\text{Unemployment} = {} & 6.1607 - 1.2716(\text{JobOpenings}) - 0.1287(\text{GrowthrateGDP}) \\ & + 2.3069(\text{Inflationrate}) - 0.5224(\text{FedFundsrate}) + 0.0175(\text{GrowthBFS})\end{aligned}$$

Where:

- JobOpenings = Number of job openings
- GrowthrateGDP = Growth rate of GDP
- Inflationrate = Inflation rate

- FedFundsrate = Federal Funds rate
- GrowthBFS = Growth in Business Formation Services (BFS)

2 Data and Forecasts

Below is the table of data and forecasted unemployment for each month:

Month	Unemployment Forecast	Job Openings	Growth rate GDP	Inflation rate	Fe
1	5.04	5.1	1.8	3.44	
2	4.62	5.1	1.8	2.82	
3	2.18	5.1	1.8	2.2	
4	4.95	4.8	1.8	2.8	
5	3.75	4.8	1.9	2.44	
6	3.8	4.8	1.9	2.24	
7	2.51	5	1.9	1.8	
8	2.95	5	1.9	1.99	
9	3.13	5	1.5	1.98	
10	3.48	4.6	1.5	1.91	
11	3.15	4.6	1.5	1.77	
12	2.75	4.6	1.5	1.6	

All the Recent Trends of Indicators

December 9, 2024

1 What we have here...

Looking at the latest trends of key indicators helps us gain a clearer understanding of how our data is evolving. By diving into these trends, we can spot changes, recognize new patterns, and get a better sense of what's driving the results. This section takes a closer look at how these indicators have shifted over time and what they can tell us about where things are headed.

2 Charts and Graphs

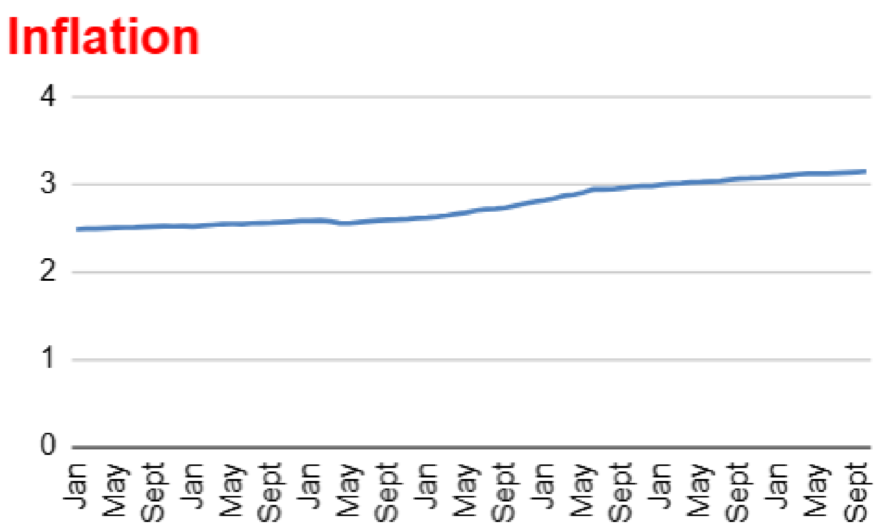

Figure 1: Recent Inflation

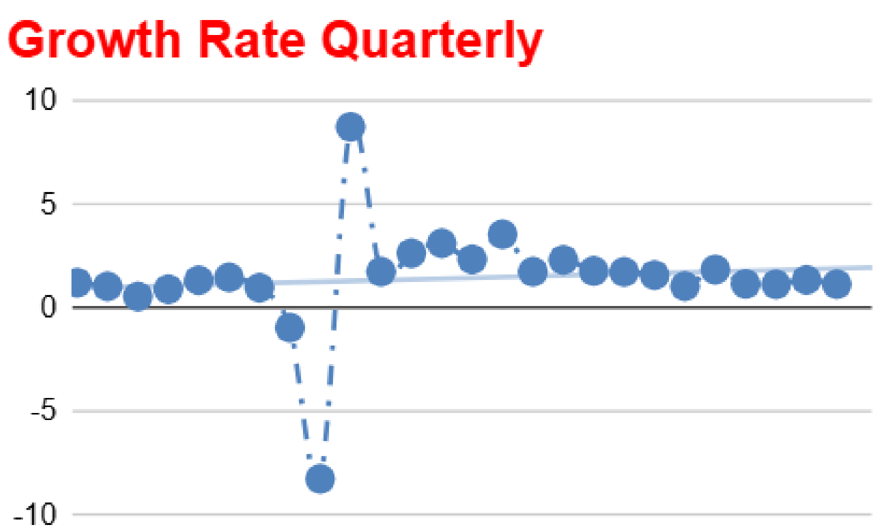

Figure 2: Recent Quarterly Growth Rate

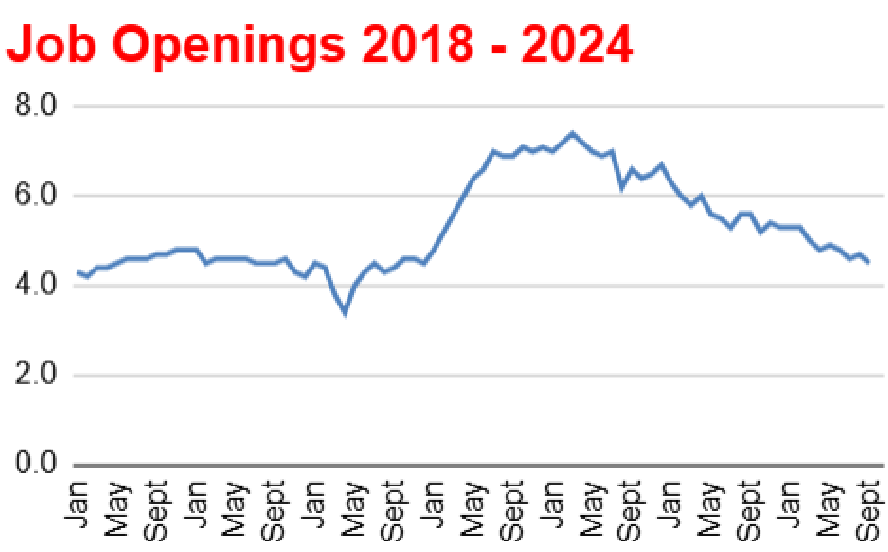

Figure 3: Recent Job Openings Trend

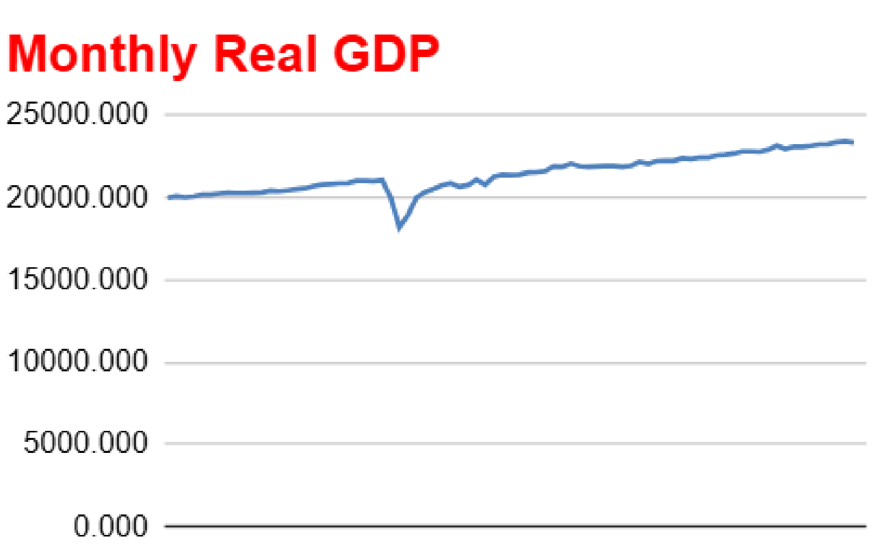

Figure 4: Monthly GDP Trend

Figure 5: Recent New Business Formations

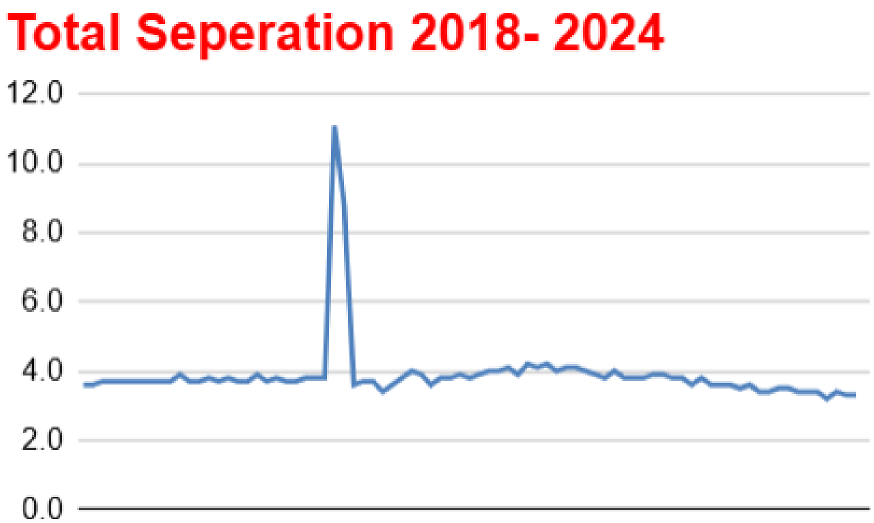

Figure 6: Recent Seperations

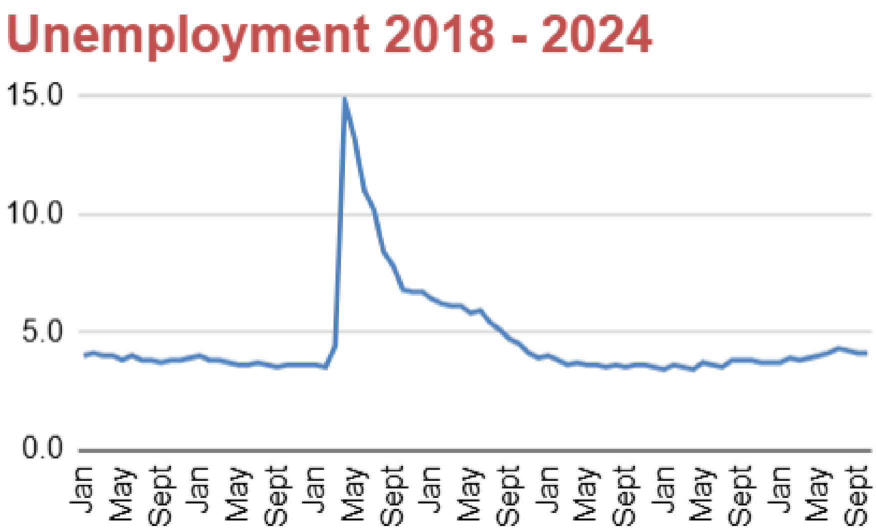

Figure 7: Recent Unemployment

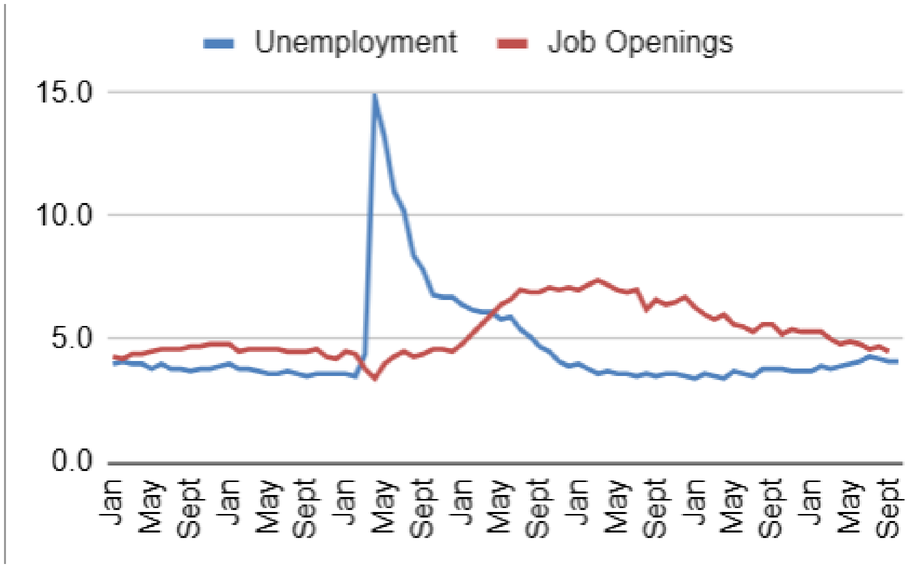

Figure 8: Unemployment and Job Openings

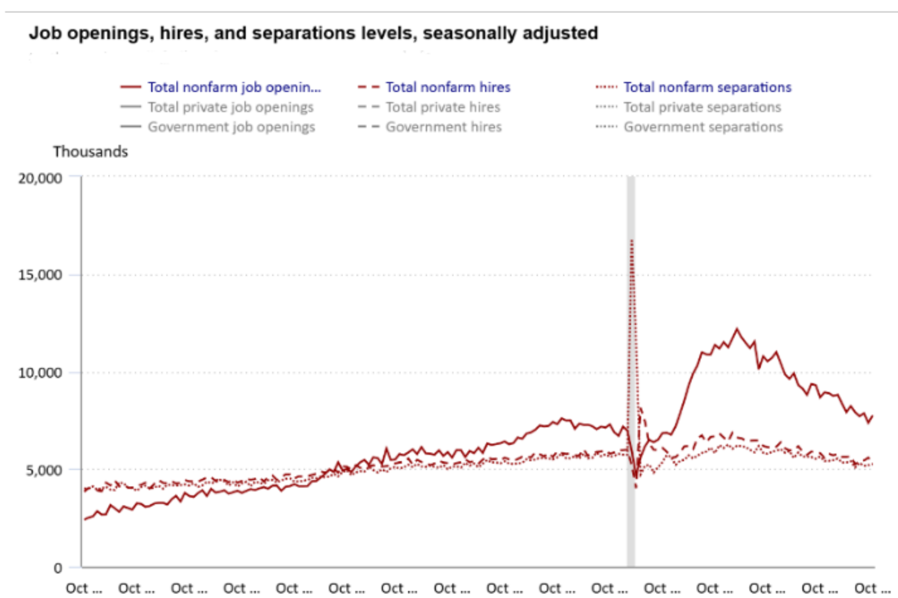

Figure 9: Hires, Seperations and Openings

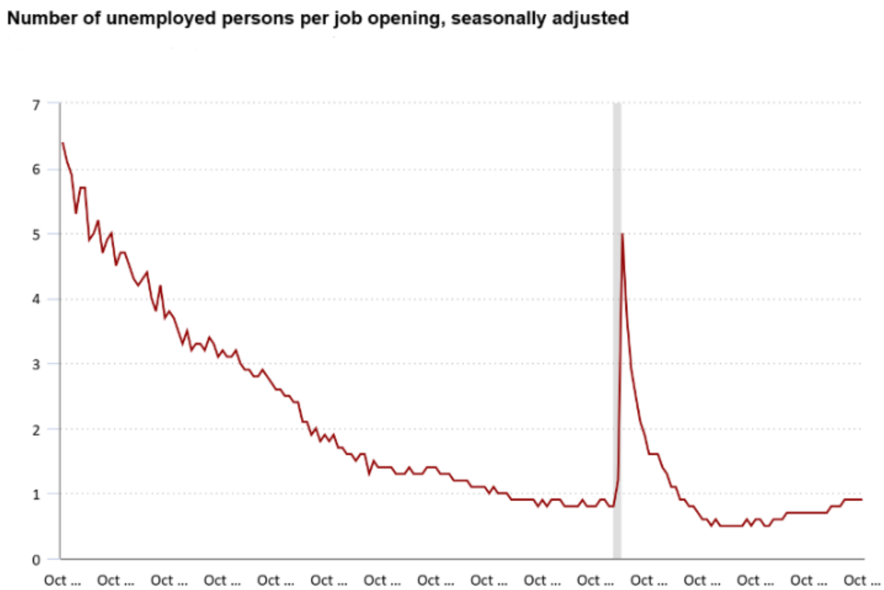

Figure 10: Unemployment-Openings Ratio

Charts, including Sectors

December 9, 2024

1 What we have here...

I rely on these graphs to deepen my qualitative understanding of the model results. They help me spot emerging trends and build an intuitive sense of what outcomes to expect. By visualizing the data this way, I can better anticipate patterns and make more informed decisions moving forward. These are the graphs that were seen to have a significant impact on unemployment.

2 Charts and Graphs

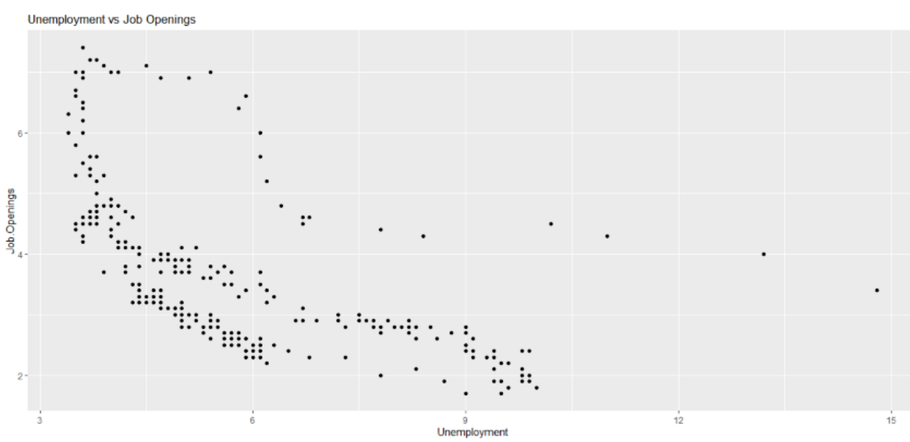

Figure 1: Beveridge Curve

Figure 2: Recent GDP Growth

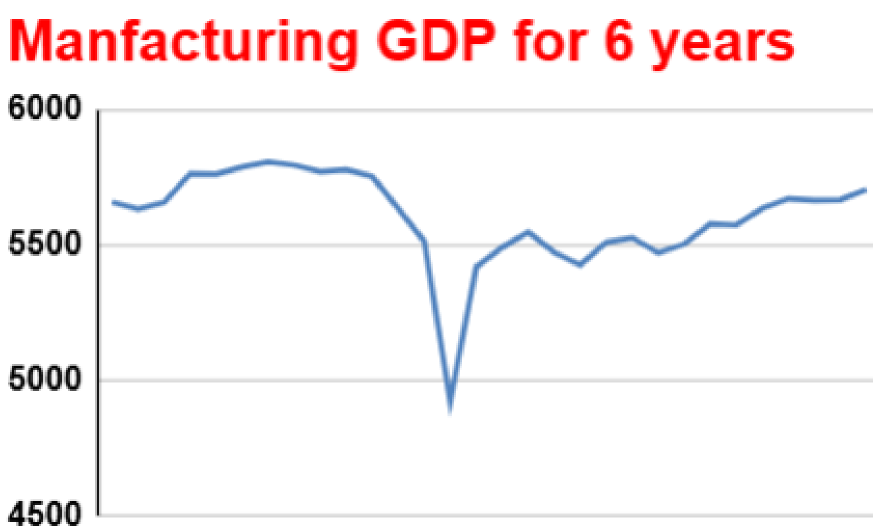

Figure 3: Manufacturing GDP

Figure 4: Education Growth

Figure 5: Agri Growth

Figure 6: Retail Sector Growth

Figure 7: Construction Sector Growth